

# EEE Final Presentation

Mustafa's Team

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# Roadmap

1. Motivation

2. Research  
Overview

3. Methodology

4. Evan's Work:  
Data Engineering

5. Sheina's Work:  
Building Context

6. Nitya's Work:  
Further Studies

7. Next Steps

8. Final Thoughts

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# Why does air pollution matter?

99%

of the world's  
population is  
affected by air  
pollution

>7.9M

premature deaths  
annually attributed  
to air pollution

4.9M

deaths in 2023 are  
linked to PM 2.5  
data

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# PM 10

All particles with aerodynamic diameter <10 **micro-meter**

**Examples:**

- Dust
- Pollen
- Mold

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# PM 2.5

All particles with aerodynamic diameter <2.5 **micro-meter**

**Examples:**

- Combustion Particles
- Organic Compounds
- Metals

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# PM 1

All particles with aerodynamic diameter <1 **micro-meter**

**Examples:**

- Soot Particles
  - Viruses
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## Our Research Project

Air pollution stations only measure the total amount of pollution, like PM2.5

**Problem Statement:** No system exists to globally estimate the contribution of dust to PM2.5 pollution

- Only available on the country or regional level in specific areas
- Also limited by the timeframes that air pollution data was collected

**Research Proposal:** Using a machine learning approach to estimate the contribution of dust events to PM2.5



Estimates can be used to measure the impact of dust pollution on human health concerns and mortality

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# Methodology - A Two Stage Design

## Pre-Processing:

Dust Days vs. Non-Dust Days categorized by dust aerosol optical depth

## Stage 1:

### Counterfactual Data

Training: Non-dust data

Input: Dust Data

Output: Counterfactual PM 2.5 Readings

Dust PM 2.5 =

True - Counterfactual

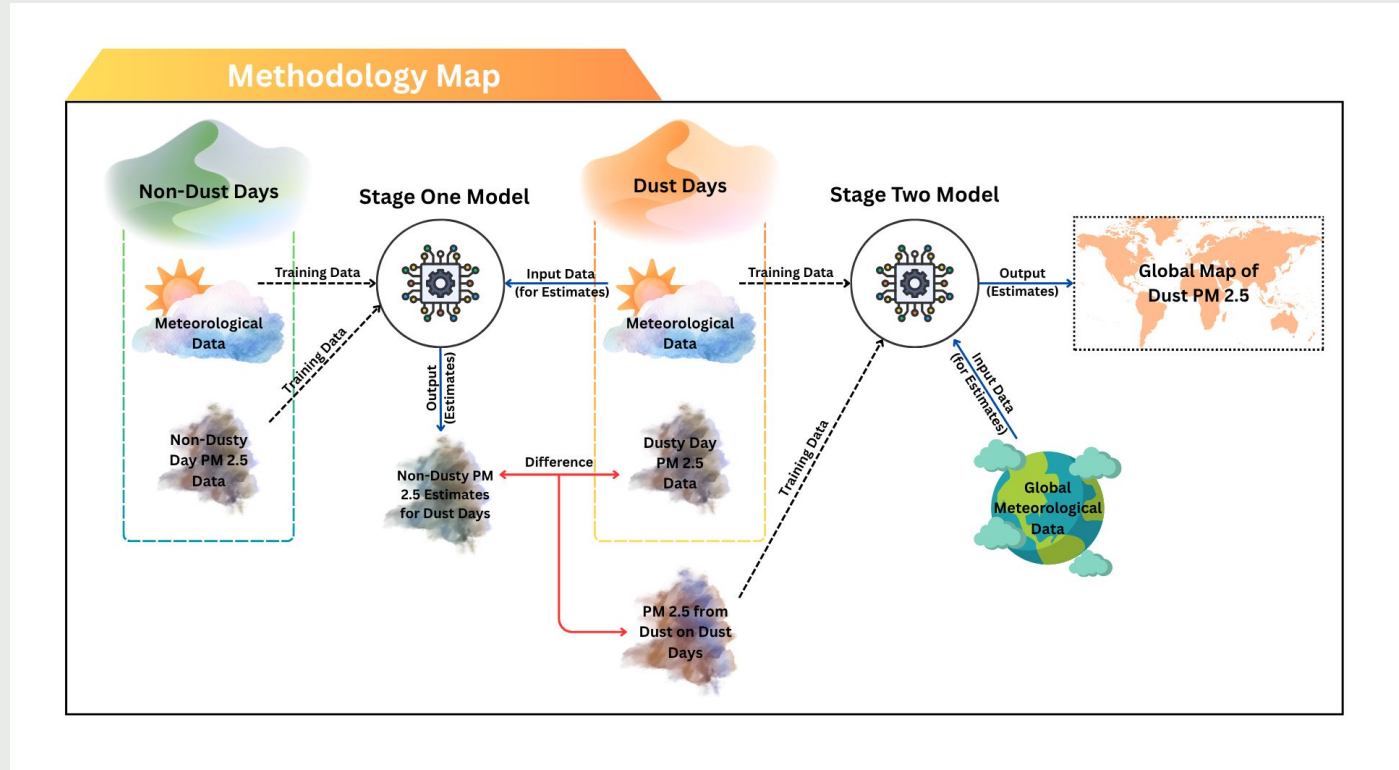
## Stage 2:

### A Global Map

Training: Dust PM 2.5

Input: Local Conditions

Output: Dust PM 2.5



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# Working with Geospatial Data



## Understanding the Data we Have

| Category | PM 2.5? | PM 10? |
|----------|---------|--------|
| Data set | Yes     | No     |
| Agency   | No      | Yes    |

## Combining our Data (Raster Extraction)

1. Get the monitor's location
2. Find which raster cell it's in
3. Extract the value of that cell for each day in the dataset
4. Repeat steps 1-3 for each monitor
5. Repeat steps 1-4 for each environmental variable of interest

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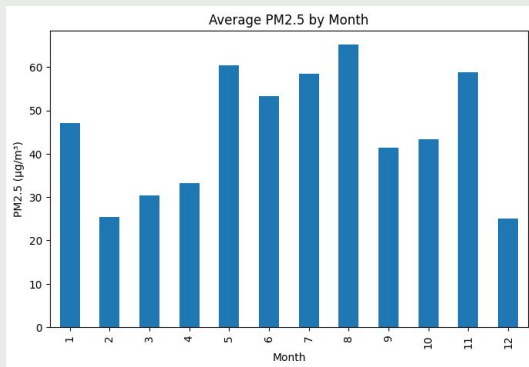
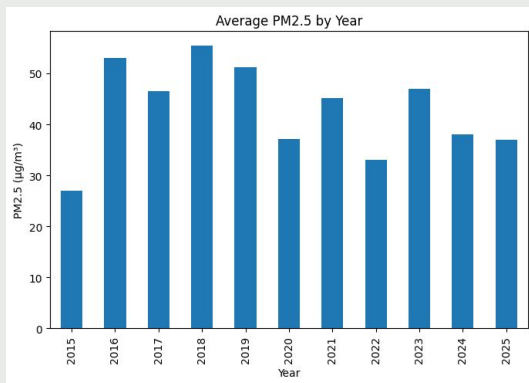
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# Connecting Data to Context



## Dry season amplifies pollution

Jakarta's pollution gets worse during the dry season because rainfall drops and the air cannot disperse particles as well. That allows emissions from transport, industry, construction, and waste burning to build up more easily.

## Some years are worse than others

Certain years spike more sharply when weather conditions intensify the problem. El Niño can extend the dry season, and regional fires can add smoke that pushes pollution even higher.

## What the chart really shows

It shows the interaction between urban growth, energy use, transport dependence, climate variability, and regional environmental risk.

## Why it matters

Poor air quality affects health, productivity, and long-term economic performance. It also increases pressure on policymakers to improve clean transport, control industrial emissions, and strengthen air-quality planning.

Other **specific events** that could play a factor in the PM2.5 spikes:

### Mid-2023 pollution crisis

Jakarta saw a widely reported pollution emergency in 2023, with the government convening cabinet-level responses and even temporary measures like water spraying and shutting a suspected charcoal factory.

### 2025 fire season

Indonesia experienced a surge in hotspots in 2025, which sent smoke across parts of the region and likely reinforced poor air conditions during the dry months.

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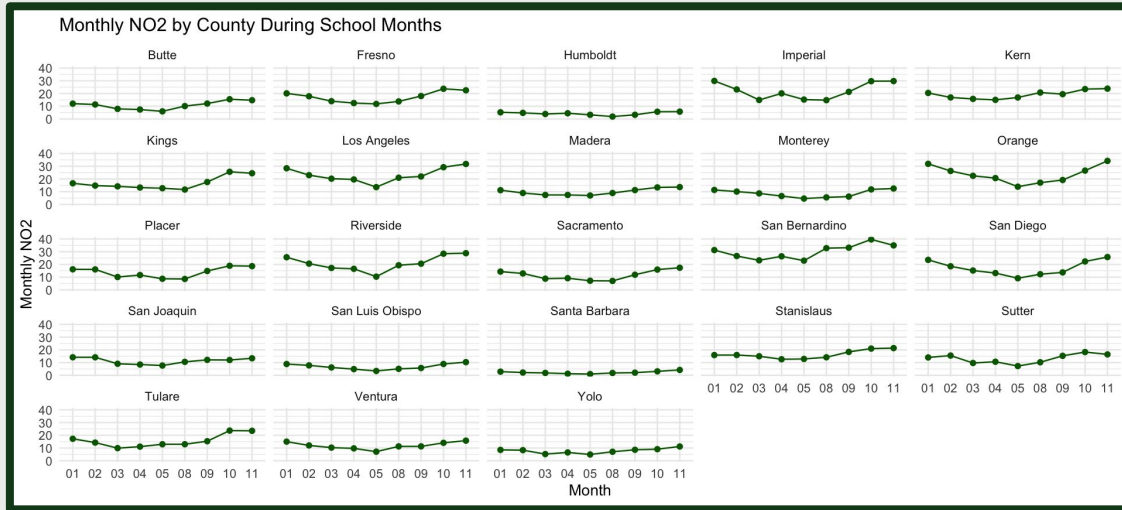
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# Another Application: Air Pollution and Attendance-Based Funding

Examining the relationship between PM 2.5 and NO2 levels in California and public school absenteeism, and how this affects funding allocations



- Downloaded daily NO2 in California by county
- Downloaded yearly chronic absenteeism data by county
- Narrowed down NO2 data to months school was in session
- Next: Looking at the days/months where AQI was above the identified threshold

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# Next Steps

Estimate  
data in  
past

Compare to  
previous  
models of  
dust  
transport

Apply data  
(to health  
outcomes,  
distribution  
of exposure,  
etc.)

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# Main Takeaway

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In research, it is important to be  skeptical and flexible.

Question your data and findings,  
and be ready to adjust



↳ course accordingly



Thank you!

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